

Engineering Probability HW 6

Timothy Pollard, pollat@rpi.edu

October 11th, 2025

(1)

Continuous to Discrete: Uniform Random Variables. Let X be a continuous uniform random variable on the interval $[0, 3]$

$$f_X(x) = \begin{cases} \frac{1}{3}, & 0 \leq x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Define a new random variable $Y = \lfloor X \rfloor$ (the smallest integer greater than or equal to X).

(a) Find the set of possible values of Y .

$$S = \{0, 1, 2\}$$

(b) Determine the PMF $P_Y(y)$ of Y .

$$P_Y(y) = \begin{cases} \frac{1}{3}, & y = 0, 1, 2, \\ 0, & \text{otherwise.} \end{cases}$$

(c) Verify that $P_Y(y)$ sums to 1.

$$P_Y(S) = P_Y(0) + P_Y(1) + P_Y(2) = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \boxed{1}$$

(d) Compute $E[Y]$ and $\text{Var}[Y]$, and compare them with $E[X]$ and $\text{Var}[X]$.

$$E[Y] = 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 2 \cdot \frac{1}{3} = \boxed{1}$$

$$E[Y^2] = 0^2 \cdot \frac{1}{3} + 1^2 \cdot \frac{1}{3} + 2^2 \cdot \frac{1}{3} = \frac{5}{3}$$

$$\text{Var}[Y] = E[Y^2] - (E[Y])^2 = \frac{5}{3} - (1)^2 = \boxed{\frac{2}{3}}$$

$$E[X] = \int_0^3 \frac{1}{3}x dx = \frac{1}{6}x^2 \Big|_0^3 = \frac{9}{6} = \boxed{1.5}$$

$$E[X^2] = \int_0^3 \frac{1}{3}x^2 dx = \frac{1}{9}x^3 \Big|_0^3 = 3$$

$$\text{Var}[X] = E[X^2] - (E[X])^2 = 3 - (1.5)^2 = \boxed{0.75}$$

Y has a 0.5 lower because it was rounded down on average 0.5. Then the variance is two thirds on x because theres two values of three values of distance one away while its evenly distributed in the continuous.

(2)

Continuous to Discrete: Exponential to Geometric. Let X be an exponential random variable with parameter $a = 0.5$, i.e.,

$$f_X(x) = \begin{cases} 0.5e^{-0.5x}, & x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Define the integer-valued random variable $K = \lceil X \rceil$.

(a) Show that $P[K = k] = e^{-0.5(k-1)}(1 - e^{-0.5})$, for $k = 1, 2, 3, \dots$

$$\begin{aligned} P[K = k] &= P[k-1 < X \leq k] = F_X(k) - F_X(k-1) \\ &= (1 - e^{-0.5k}) - (1 - e^{-0.5(k-1)}) \\ &= e^{-0.5(k-1)} - e^{-0.5k} \\ &= e^{-0.5(k-1)}(1 - e^{-0.5}) \end{aligned}$$

(b) Identify the parameter p of the geometric random variable K and verify that $p = 1 - e^{-a}$.

$$\begin{aligned} \sum_{k=1}^{\infty} e^{-0.5(k-1)} (1 - e^{-0.5}) \\ p = 1 - e^{-0.5} \end{aligned}$$

(c) Compute $E[K]$ and compare it to $E[X] = \frac{1}{a}$. Comment on that relationship.

$$E[K] = \frac{1}{p} = \frac{1}{1 - e^{-0.5}} \approx 2.54 \quad E[X] = \frac{1}{0.5} = 2,$$

$E[K]$ is Roughly 0.5 more than K which makes sense becaus e its rounded up. Its not excat because it has a offset.

(3)

Exponential Random Variable via Variance. Let Y be an exponential random variable with variance $\text{Var}[Y] = 25$.

(a) What is the PDF of Y ?

$$25 = \frac{1}{a^2} \therefore a = \frac{1}{5}, f_Y(y) = \begin{cases} \frac{1}{5}e^{-\frac{y}{5}}, & y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

(b) What is $E[Y]$?

$$E[Y] = \frac{1}{a} = \boxed{5}$$

(c) What is $P[Y > 5]$?

$$\begin{aligned} P[Y > 5] &= 1 - F_X(5) = 1 - \int_0^5 \frac{1}{5}e^{-\frac{y}{5}} dy \\ &= 1 - \left(1 - e^{-\frac{y}{5}}\right) \Big|_0^5 \\ &= 1 - (-e^{-1} + 1) = \frac{1}{e} \approx \boxed{0.367} \end{aligned}$$

(4)

Uniform on a Symmetric Interval. Let X be a continuous random variable uniformly distributed over $[-5, 5]$.

(a) Determine the PDF $f_X(x)$.

$$f_X(x) = \begin{cases} \frac{1}{10}, & -5 \leq x \leq 5, \\ 0, & \text{otherwise.} \end{cases}$$

(b) Determine the CDF $F_X(x)$.

$$F_X(x) = \begin{cases} \frac{x}{10}, & -5 \leq x \leq 5, \\ 0, & x < -5 \\ 1, & x > 5 \end{cases}$$

(c) Compute $E[X]$.

$$E[X] = \int_{-5}^5 \frac{1}{10} x dx = \frac{1}{20} x^2 \Big|_{-5}^5 = \frac{(5)^2}{20} - \frac{(-5)^2}{20} = \boxed{0}$$

(d) Compute $E[X^2]$.

$$E[X^2] = \int_{-5}^5 \frac{1}{10} x^2 dx = \frac{1}{30} x^3 \Big|_{-5}^5 = \frac{(5)^3}{30} - \frac{(-5)^3}{30} = \frac{250}{30} = \boxed{\frac{25}{3}}$$

(e) Compute $E[e^{5X}]$.

$$E[e^{5X}] = \int_{-5}^5 \frac{1}{10} e^{5x} dx = \frac{1}{50} e^{5x} \Big|_{-5}^5 = \frac{e^{25}}{50} - \frac{e^{-25}}{50} \approx \boxed{1.44 \cdot 10^9}$$

(5)

Erlang Random Variable. Let X be an Erlang random variable with parameters $\lambda = 1$ and $n = 3$:

$$f_X(x) = \begin{cases} \frac{x^2 e^{-x}}{2}, & x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Verify that $f_X(x)$ is a valid PDF.

$$\int_0^\infty \frac{x^2 e^{-x}}{2} dx =$$

D	I	
$\frac{x^2}{2}$	$+$	e^{-x}
x	$-$	$-e^{-x}$
1	$+$	e^{-x}
0	$-$	$-e^{-x}$

$$= \left(-\frac{x^2}{2} e^{-x} - x e^{-x} - e^{-x} \right) \Big|_0^\infty$$

$$= -e^{-x} \left(\frac{x^2}{2} + x + 1 \right) \Big|_0^\infty$$

$$= (0) - (-1)$$

$$= 1$$

(b) Compute $E[X]$ and $\text{Var}[X]$.

$$E[X] = \int_0^\infty \frac{x^3 e^{-x}}{2} dx$$

D	I	
$\frac{x^3}{2}$	$+$	e^{-x}
$\frac{3x^2}{2}$	$-$	$-e^{-x}$
$3x$	$+$	e^{-x}
3	$-$	$-e^{-x}$
0	$-$	e^{-x}

$$E[X] = \left(-\frac{x^3}{2} e^{-x} - \frac{3x^2}{2} e^{-x} - 3x e^{-x} - 3e^{-x} \right) \Big|_0^\infty$$

$$E[X] = -e^x \left(\frac{x^3}{2} + \frac{3x^2}{2} + 3x + 3 \right) \Big|_0^\infty$$

$$E[X] = (0) - (-3)$$

$$E[X] = \frac{n}{\lambda} = \boxed{3}$$

$$\text{Var}[X] = \frac{n}{\lambda^2} = \frac{3}{1^2} = \boxed{3}$$

(c) Find $P[X > 2]$.

$$\begin{aligned} P[X > 2] &= 1 - F_X(2) \\ &= 1 - \int_0^2 \frac{x^2 e^{-x}}{2} dx \text{ (Already Solved Above)} \\ &= 1 - \left(-e^{-x} \left(\frac{x^2}{2} + x + 1 \right) \Big|_0^2 \right) \\ &= 1 - \left(-e^{-2} \left(\frac{2^2}{2} + (2) + 1 \right) - ((-1)(1)) \right) \\ &= e^{-2}(5) \approx \boxed{(0.677)} \end{aligned}$$

(6)

Gaussian Random Variable. Let X be a Gaussian random variable with mean $\mu = 50$ and standard deviation $\sigma = 10$.

- (a) Write the PDF $f_X(x)$ and define the standard normal variable $Z = \frac{X-\mu}{\sigma}$.

$$f_X(x) = \frac{1}{10\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-50}{10}\right)^2}$$

- (b) Compute $P[40 \geq X \geq 60]$ using the standard normal CDF $\Phi(z)$.

$$P[40 \geq X \geq 60] = \Phi\left(\frac{60-50}{10}\right) - \Phi\left(\frac{40-50}{10}\right) \approx 0.8413 - (1 - 0.8413) \approx \boxed{0.6826}$$

- (c) Compute $P[X > 70]$ using the Q-function, $Q(z) = 1 - \Phi(z)$.

$$P[X > 70] = Q\left(\frac{70-50}{10}\right) \approx \boxed{0.02275}$$

- (d) Let $Y = 3X - 20$. Find μ_Y, σ_Y^2 and the PDF $f_Y(y)$.

$$\mu_Y = 3(50) - 20 = 130 \quad \sigma_Y^2 = (3(10))^2 = 900$$

$$f_Y(y) = \frac{1}{30\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{y-130}{30}\right)^2}$$